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Professional Self-Assessment

Throughout the Computer Science program at Southern New Hampshire University, I have built a foundation that combines software development skills with a strong focus on delivering reliable, decision supporting solutions. Completing CS 499 and developing this ePortfolio helped me shift from simply finishing assignments to thinking like a professional, with more attention on maintainability, scalability, data integrity, and clear communication. After graduation, I plan to apply for Data Analyst, Business Intelligence Analyst, Scrum Master, and Product Manager roles. While these roles look different day to day, they all rely on understanding systems and data well enough to help teams make good decisions and deliver outcomes that stakeholders can trust.

A major strength I developed across the program is the ability to collaborate and communicate with different audiences. Through iterative feedback in coursework and repeated revisions, I practiced translating technical work into clear explanations of what a system does, why a design decision matters, and what trade offs come with it. In CS 499, this became even more important because each enhancement required both working code and a narrative that could be understood by technical and nontechnical readers. This communication skill supports every role I am pursuing. Analysts need to explain findings clearly, Scrum Masters need to keep teams aligned, and Product Managers need to connect priorities to outcomes in a way that makes sense to stakeholders.

This portfolio highlights my growth in software engineering, where I improved the structure and maintainability of an existing backend module by refactoring responsibilities into clearer components. I focused on separation of concerns, reliability, and making future changes easier to implement without breaking existing functionality. This kind of modular thinking is valuable beyond software development. It also supports analytics and product work because data pipelines, dashboards, and services evolve over time, and weak structure quickly becomes a bottleneck.

My work in algorithms and data structures demonstrates how I evaluate efficiency and correctness when designing computing solutions. I improved scalability by selecting data structures that reduce unnecessary searching and by implementing stronger validation logic tied to real constraints in the data. Beyond performance, this work reflects a professional habit of choosing solutions based on the problem context and explaining trade offs clearly. For analyst roles, this mindset translates into efficient data processing and dependable results. For Scrum Master and Product Manager roles, it supports better estimation, prioritization, and risk management because technical complexity and constraints are understood rather than guessed.

My database enhancement demonstrates database centered skills that support analytics and reliable decision making. I implemented improvements that strengthen data integrity, improve query performance, and support analysis ready outputs. This includes practices such as enforcing uniqueness and optimizing retrieval patterns so systems scale while maintaining trust in the data. These skills align strongly with Data Analyst and Business Intelligence roles, but they also support product and delivery work by ensuring the underlying systems remain accurate, performant, and dependable for stakeholders.

A security mindset runs throughout my approach. Across my work, I focused on protecting sensitive information, preventing avoidable integrity issues, and designing systems that reduce risk. Even when a project is not explicitly security focused, I have learned to anticipate where vulnerabilities can emerge, such as hardcoded secrets, weak validation, or overreliance on application only checks, and to mitigate those risks through better design and database controls. This matters in every role I am applying for because trust and quality are part of delivering a strong outcome.

How this ePortfolio supports multiple roles:

Data Analyst and Business Intelligence Analyst: This portfolio demonstrates data integrity, query efficiency, structured problem solving, and clear communication of results and trade offs.

Scrum Master: This portfolio demonstrates disciplined planning, iterative improvement, clear written communication, and the ability to translate technical work into stakeholder friendly language.

Product Manager: This portfolio demonstrates understanding of system design and constraints, decision making based on trade offs, and delivering value through structured, measurable enhancements.

Together, the artifacts in this ePortfolio tell a connected story about growth and delivery. The software engineering enhancement shows maintainable design and reliability. The algorithms and data structures enhancement shows performance reasoning and correctness validation. The database enhancement shows integrity and efficient access to data for analysis. Across all three, I demonstrate the ability to evaluate an existing solution, identify limitations, implement

improvements with clear reasoning, and communicate those decisions effectively. These are skills that translate directly to analytics, product, and delivery focused roles.

Key Skills Highlighted

- Data reasoning and decision support through reliable systems
- Algorithmic thinking focused on efficiency, validation, and trade offs
- Database concepts including integrity, indexing, optimization, and analysis friendly querying
- Technical communication for varied audiences
- Process mindset that emphasizes iteration, prioritization, and quality focused delivery